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Bihar Engineering University, Patna

B.Tech 8th Semester Exam., 2022

Code: 100813 | ***Time: 03 Hours***

Semester: VIII | ***Subject: BIG DATA ANALYTICS*** | ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Answer/Choose the correct option of the following (any seven) : **[2 x 7 = 14]**

(a) Which of the following is not a characteristic of Big Data?

(i) Volume
(ii) Variability
(iii) Variety
(iv) Velocity

(b) Which of the following terms is used to denote the small subsets of a large file created by HDFS?

(i) NameNode
(ii) DataNode
(iii) Blocks
(iv) Namespace

(c) Technology that manages traffic between virtual and physical machines is known as

(i) MapReduce
(ii) Hadoop
(iii) Hypervisors
(iv) Virtualization

(d) What happens, if the mapper output does not match the reducer input?

(i) The Hadoop API will convert the data to the type needed by the reducer
(ii) Data input/output inconsistency is not permitted. Therefore, a preliminary validation check is executed prior to the full execution of the job to ensure consistency
(iii) The Java compiler will report an error during compilation, but the job will complete with exceptions
(iv) A real-time exception will be thrown, and the MapReduce job will fail

(e) Which of the following is YARN scheduler?

(i) Fair Scheduler
(ii) Quartz Scheduler
(iii) Huge Data
(iv) None of the above

(f) The CREATE statement in Hive is related to

(i) DDL statements
(ii) DML statements
(iii) Session control statements
(iv) Embedded SQL statements

(g) Which mode of Pig is also known as the Hadoop mode?

(i) Local mode
(ii) MapReduce mode
(iii) Global mode
(iv) Universal mode

(h) Which of the following is a part of the unsupervised learning concept?

(i) Data mining
(ii) Recommendation
(iii) Bar-graph representation
(iv) K-means

(i) Which of the following analytical tools is mostly used in social sciences, marketing and healthcare?

(i) R
(ii) SPSS
(iii) JMP
(iv) Spotfire

(j) Write a short note on the FileInputFormat class.

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Q.2 Solve both questions : **[7 x 2 = 14]**

(a) What is big data? Discuss the key features of big data

(b) How do big data analytic techniques reform the modern business industry? Discuss in detail.

Q.3 Solve both questions : **[7 x 2 = 14]**

(a) Differentiate between predictive analytics and prescriptive analytics with suitable example

(b) What is the difference between data analysis and data reporting?

Q.4 **[14]**

Discuss the modern big data analysis tools to detect malicious activities on social media platform.

Q.5 **[14]**

List the important feature of Hadoop. Illustrate the computing model of Hadoop with neat labelled diagram.

Q.6 Solve both questions : **[7 x 2 = 14]**

(a) Explain how matrix multiplication is performed using MapReduce algorithm.

(b) What is multivariate analysis? Discuss advantages and disadvantages of multivariate analysis.

Q.7 **[14]**

Draw a neat labelled diagram of HDFS architecture. Briefly discuss each component of HDFS architecture.

Q.8 Solve both questions : **[7 x 2 = 14]**

(a) Discuss the strength and weakness of density-based clustering algorithm.

(b) Explain the common procedure followed for extracting reliable sample from a stream.

Q.9 Solve both questions : **[7 x 2 = 14]**

(a) Compare relational databases with NoSQL databases.

(b) What are various statements used in flow of data processing in Pig Latin?

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